



Attack chain contraction and prediction using Markov model and LSTM on MITRE ATT&CK data: a thesis in Data Science.

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University of Massachusetts Dartmouth

Attack Chain Contraction and Prediction Using Markov Model and
LSTM on MITRE ATT&CK Data

A Thesis in

Data Science

by

Ruksar Rafique Lukade

Submitted in Partial Fulfillment of the

Requirements for the Degree of

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Abstract

Attack Chain Contraction and Prediction Using Markov Model and LSTM on MITRE ATT&CK Data

by Ruksar Rafique Lukade

As cyber threats grow in complexity, understanding how adversaries operate and predicting their next moves has become essential for proactive defense. This thesis presents a hybrid modeling system that combines logic-driven simulation with machine learning–based sequence prediction to anticipate the progression of cyberattack chains. Using structured data from the MITRE ATT&CK framework, we parse techniques, tactics, threat groups, campaigns, and inter-relationships to generate realistic, directed attack chains through both randomized simulation and campaign-informed permutations.

To simulate how attackers might move through systems, we train a first-order Markov Chain model that captures transition probabilities between techniques and supports probabilistic multi-step path generation. In comparison, a Long Short-Term Memory (LSTM) neural network learns to predict context-aware next-step techniques based on full sequence history, capturing deeper temporal and semantic patterns. We also introduce a geometric mean–based scoring method to evaluate the risk and coherence of each predicted path, categorizing them into low, medium, or high-risk levels. To reduce noise and improve interpretability, we apply a Chain Contraction algorithm that compresses redundant or semantically similar steps, producing cleaner and more meaningful representations of attacker behavior. An interactive interface allows users to select a starting technique from a dropdown menu and explore simulated attack paths in real time, complete with tabulated probabilities and visual graph representations.

We evaluate both models by comparing their predictions to real-world campaigns such as Operation Ghost (C0023) and SolarWinds (C0024) from the APT29 threat group (G0016), using [STIX-based datasets from online repositories](#) from the ATT&CK. This work demonstrates how structured threat intelligence and generative modeling can work together to support red teaming, threat hunting, and campaign attribution. It provides hands-on tools for analysts to simulate, visualize, and compare attacker behavior in a dynamic, data-driven environment.

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Chapter 1: Introduction

1.1 Problem Statement

In today's cybersecurity landscape, organizations are increasingly confronted with sophisticated, stealthy, and coordinated threats posed by Advanced Persistent Threats (APTs) [1]. These adversaries are capable of executing multi-stage attacks that involve varied combinations of tactics and techniques, often spanning days, weeks, or even months [2] [3]. Such campaigns are not only complex but also adaptive, rendering traditional defense systems reactive and insufficient [2].

The MITRE ATT&CK framework has emerged as a standardized knowledge base to characterize adversary behaviors [4]. It defines a matrix of tactics (goals) and techniques (actions) that attackers may employ at each stage of an intrusion, from reconnaissance to impact [4]. However, while ATT&CK is widely adopted for post-hoc analysis, its predictive capabilities remain underdeveloped [5] [6] [7]. Current uses of the framework tend to rely on static graphs or predefined paths, limiting their ability to anticipate how an attacker might progress given a partial sequence of observed behaviors [5] [7]. As attackers innovate, defenders are left with minimal tooling for forecasting adversarial behavior, particularly in scenarios where defenders only observe part of an attack chain [6] [7].

The challenge is twofold. First, there is a need for a modeling framework that can generate and predict adversary behavior chains dynamically, not just replay known attack paths [8] [9] [10]. Second, there is a need to improve the interpretability of such models, particularly when automated techniques produce long, noisy, or redundant outputs that overwhelm analysts [5].

In practice, security analysts need systems that not only predict potential next steps but also help them make sense of which attack paths matter most, how those predictions relate to known campaigns, and what actions they should take. Given the scale and complexity of modern threat datasets, machine learning and structured simulations offer promising avenues for bridging this gap.

This thesis addresses these challenges by proposing a hybrid AI-driven framework for dynamic attack chain prediction and simulation, grounded in the MITRE ATT&CK ontology. The system combines probabilistic modeling (Markov Chains) with deep learning (LSTM networks) and introduces a chain contraction technique to simplify results. It further incorporates campaign-based simulation using real-world STIX relationships and evaluates predictions against known campaigns like Operation Ghost (C0023) and SolarWinds (C0024) of the APT29 (G0016) threat group [1].

In summary, this work aims to solve two core problems:

1. Accurate, dynamic prediction of adversary behavior chains using structured threat intelligence and machine learning, and
2. Improved usability and interpretability of such predictions through visualization, risk classification, and semantic compression.

These capabilities are essential for proactive cyber defense, red teaming, and threat hunting in modern security environments.

1.2 Method Overview and Distinction from Existing Solutions

This thesis proposes a hybrid modeling framework to simulate, predict, and simplify cyberattack chains by leveraging structured knowledge from the MITRE ATT&CK framework [4] [5]. The system enhances both the predictive accuracy and interpretability of adversarial behavior sequences by combining first-order Markov Chains [11] [12] with Long Short-Term Memory (LSTM) neural networks [13] [9] [10] and applying a chain contraction algorithm. Additionally, it incorporates campaign-grounded chain generation using real-world ATT&CK adversary data [1], risk scoring using geometric mean probabilities [6], and visual analytics via interactive widgets. These components enable analysts to simulate realistic attack sequences, identify likely next steps, and compress chains into concise, semantically coherent summaries.

The framework comprises five interdependent modules. Markov Chain-Based Attack Simulation and Risk Classification, LSTM-Based Sequence Prediction, Campaign-Informed Chain

Synthesis, Chain Contraction Algorithm, and an Interactive Interface module, developed to test and evaluate the outputs of this thesis.

1.2.1 Core System Components

1.2.1.1 Markov Chain–Based Attack Simulation and Risk Classification

The system constructs a first-order Markov Chain model based on transitions observed in both simulated and campaign-extracted attack chains [11] [12]. Transition counts are aggregated to produce a transition probability matrix. Random walks constrained by tactic ordering, loop avoidance, and memory of previous transitions simulate plausible attacker movement. For each generated path, the geometric mean of transition probabilities is computed to score the path’s consistency and risk level (Low, Medium, High) [6]. A transition tree visualization is generated using NetworkX to display top branches originating from a chosen technique.

1.2.1.2 LSTM-Based Sequence Prediction Using PyTorch

A PyTorch-based LSTM model is trained on tokenized and padded sequences of technique chains [13]. Techniques are integer-encoded, embedded in a 64-dimensional space, and passed through a single-layer LSTM with 128 hidden units. A fully connected layer followed by SoftMax activation predicts the next likely technique [9] [10]. The model is trained using Adam optimizer and cross-entropy loss for 10 epochs. Inference is recursive, generating top k multi-step paths from an input prefix, pruned using confidence thresholds and loop prevention. This allows the model to predict non-obvious, long-range transitions across tactics.

1.2.1.3 Campaign-Informed Chain Synthesis

To complement simulated data, the framework synthesizes realistic chains from APT29 ATT&CK campaigns (e.g., C0023, C0024) [4]. Techniques linked to each campaign are grouped by primary tactic, permutations are computed within each group, and tactic-preserving Cartesian products generate diverse chain samples. Only sequences above a minimum length threshold are retained. These paths serve as both LSTM training data and ground-truth evaluation sets [7].

1.2.1.4 Chain Contraction Algorithm for Sequence Compression

To reduce analyst fatigue and enhance clarity, predicted chains are post-processed using a Chain Contraction algorithm [5]. This component eliminates repeated techniques, filters low-probability transitions (under a configurable confidence threshold), and prunes redundant segments based on entropy analysis. Tactic-based clustering is maintained to preserve the strategic structure of attacks, resulting in a shorter, high-utility representation of adversary movement.

1.2.1.5 Interactive Interface and Evaluation Metrics

A Jupyter Notebook interface built with ipywidgets allows users to select a starting technique via dropdown and trigger the full simulation pipeline. Outputs include tabulated Markov and LSTM predictions, interactive graph visualizations of attack paths, and statistical comparisons of both models. Evaluation involves overlap matching against known techniques from Operation Ghost and SolarWinds [1]. Summary statistics include prediction accuracy, chain overlap, chain length reduction, and model-specific path divergence.

1.2.2 Distinction from Existing Solutions

This framework addresses critical limitations of prior adversary modeling approaches and introduces several novel capabilities:

- **Hybrid Symbolic-Neural Modeling:** By combining first-order Markov Chains [11] [12] with LSTM-based sequence modeling [13] [9] [10], the system balances interpretability and predictive depth, capturing both frequent short-range transitions and long-range semantic patterns.
- **Campaign-Grounded Simulation:** Unlike prior methods that use synthetic or randomly sampled sequences, this framework generates chains directly from real-world MITRE ATT&CK campaigns [1] using tactic-grouped permutations. This ensures realism and contextual accuracy [7].

- **Risk-Aware Path Scoring:** Chains generated by the Markov model are evaluated using geometric mean-based risk scoring, enabling prioritization of high-confidence, stable attack paths [6].
- **Semantic Chain Contraction:** A unique contraction algorithm filters low-confidence, redundant, and low-entropy transitions, producing concise, interpretable chains while preserving adversarial intent and tactic order [5].
- **Interactive, Analyst-Centric Design:** The system includes a Jupyter-based UI with dropdowns, tabulated outputs, and graph visualizations. This makes it practical for real-world use in SOCs, red teaming, and threat hunting [8].
- **Proactive Forecasting and Evaluation:** Beyond post-incident detection, the system enables proactive “what-if” simulations and compares predictions against real campaigns (C0023, C0024) from groups (APT29), supporting strategic threat anticipation [1].
- **Adaptability to New Threats:** Built on the MITRE ATT&CK framework [4], the system dynamically updates as new techniques, tactics, and relationships emerge, maintaining relevance over time [14].

This thesis contributes to a scalable, interpretable, and AI-enhanced modeling system for cyberattack simulation and prediction. By combining statistical transitions, deep learning inference, and semantic contraction, the framework supports both automated analysis and human decision-making. It serves as a practical tool for defenders seeking to understand, anticipate, and counteract adversarial behavior in dynamic threat environments.

1.3 Findings

The thesis presents a hybrid probabilistic framework that effectively models and simplifies cyberattack chains through the integration of first-order Markov Chains and Long Short-Term Memory (LSTM) neural networks. The evaluation of this framework reveals several significant findings.

1.3.1 Improved Prediction Accuracy

local transition strengths of Markov models [11] [12] with the long-range dependency modeling of LSTM networks [13] [9]. The LSTM was trained on both simulated attack sequences and real-

world campaign-informed chains [1], enabling it to generalize across diverse attack scenarios. Rather than using traditional metrics like MSE, it relied on top K prediction quality, semantic alignment, and overlapping with known adversary techniques, particularly in multi-tactic scenarios [10].

1.3.2 Reduction in Chain Complexity

The Chain Contraction algorithm played a critical role in simplifying complex multi-step attack sequences. By removing repeated, low-confidence, and low-entropy transitions while preserving tactic order, the algorithm reduced chain length by up to 40%, significantly improving interpretability [5]. This simplification enables security analysts to focus on strategically relevant transitions and enhances decision-making under time constraints.

1.3.3 Validation Against Real-World Campaigns

The framework was validated against adversary campaigns documented in the MITRE ATT&CK framework, including Operation Ghost (C0023) and the SolarWinds (C0024) incident, as well as the threat group APT29 (G0016) [1]. The system successfully matched predicted and contracted techniques with those found in campaign STIX relationships [15]. Evaluation included overlap scoring and technique match rate, confirming that the model captures realistic adversarial behavior patterns.

1.3.4 Interactive Visualization Tools

A Jupyter-based user interface was developed to allow users to explore multi-step simulated attack paths. Users can select starting techniques from a dropdown and view predicted sequences generated by both Markov and LSTM models. Each path is annotated with tactics, transition probabilities, risk scores, and contraction decisions, presented through tabulated outputs and graph-based visualizations using NetworkX and Matplotlib [8]. This tool enhances usability and supports real-time exploration analysis.

1.3.5 Scalability and Adaptability

The framework is designed for long-term adaptability. Because it is grounded in the MITRE ATT&CK knowledge base [4], it can easily be updated as new techniques, tactics, and relationships are released. The modular codebase supports future retraining, retriggering simulations, or ingesting newer STIX datasets [15], making the system scalable and sustainable for evolving cyber threat landscapes [14].

1.4 Main Impact

The findings from this thesis have significant implications for various stakeholders in the cybersecurity domain.

1.4.1 Cybersecurity Analysts and Red Teams

The proposed hybrid modeling framework provides a practical tool for simulating and visualizing adversary behavior using real-world tactics and techniques [4]. By allowing analysts to explore multi-step attack paths and compare predictions from both Markov and LSTM models, the system enhances situational awareness. The Chain Contraction mechanism improves the clarity of complex attack chains [5], enabling analysts to focus on high-priority transitions. This facilitates more efficient threat hunting, adversary emulation, and incident response planning.

1.4.2 Organizations and Security Operations Teams

Security teams can leverage the framework to simulate realistic adversary scenarios and evaluate the robustness of existing controls [2]. By forecasting possible attack paths starting from various techniques, organizations can adopt a proactive posture in identifying exposure areas. The framework aids in training exercises, tabletop simulations, and security readiness assessments against Advanced Persistent Threats (APTs) [3]. While it does not identify system-level vulnerabilities directly, it enables defenders to understand likely adversarial strategies and adjust defenses accordingly.

1.4.3 Researchers and the Academic Community

This work contributes to ongoing research in machine learning for cybersecurity [9] [10], threat modeling [11], and ATT&CK-driven simulation [5]. By combining symbolic reasoning with deep sequence prediction [13], the framework bridges structured threat intelligence [4] with generative modeling [10]. It offers a reproducible base for further study into multi-model prediction, adversary intent modeling, and campaign-level analytics [7], encouraging future innovations in both academia and industry.

The hybrid probabilistic framework not only addresses the challenges of modeling and predicting cyberattack chains [6], but also provides interpretable, actionable tools for cyber defense stakeholders, enabling proactive mitigation and deeper adversary understanding.

The subsequent chapters of this thesis are structured to provide a comprehensive exploration of the hybrid modeling framework and its application to cyberattack prediction. Chapter 2 presents a literature review, examining existing approaches to modeling adversarial behavior, including graph-based simulations and machine learning models, and outlines their limitations in dynamic, real-world scenarios. Chapter 3 details the methodology, including data extraction and parsing from the MITRE ATT&CK framework [4], the construction of the Markov transition model [11], the design and training of the LSTM neural network [9] [11], the generation of campaign-informed attack chains [1], and the implementation of the Chain Contraction algorithm [5].

Chapter 4 describes the experimental setup, including how simulated and real-world campaign chains were used to generate predictions. It also explains the evaluation techniques used to compare model outputs with documented campaigns and threat groups, specifically through technique overlap analysis, match scoring, and risk classification of predicted paths [6]. Chapter 5 discusses the results, highlighting the predictive performance of the hybrid system [9], reductions in chain complexity [5], and the alignment with real-world adversary behavior [7].

Finally, Chapter 6 concludes the thesis by summarizing key findings, outlining the practical implications of the work for analysts and threat hunters, and suggesting avenues for future research, including enhanced multi-model fusion [10] and real-time integration into security operations [6].

Chapter 2: Related Work

2.1 What Is Known: Forecasting and Understanding Adversarial Behavior

As cybersecurity threats continue to evolve in complexity, understanding how adversaries behave and forecasting their next moves have emerged as crucial goals in cyber defense. Adversarial behavior modeling is no longer confined to signature-based detection or post-incident forensics; instead, the focus is shifting toward proactive prediction, dynamic simulation, and strategic anticipation. This section reviews the state-of-the-art in forecasting adversarial behavior by examining conceptual foundations, structured knowledge bases like MITRE ATT&CK, probabilistic models, deep learning-based sequence models, graph-based learning methods, and adversary simulation tools.

2.1.1 Conceptual Frameworks for Modeling Adversary Behavior

One of the foundational contributions to adversarial behavior modeling is the Cyber Kill Chain, proposed by Hutchins et al. [2]. This model breaks down an intrusion into seven high-level phases: Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command and Control (C2), and Actions on Objectives. While widely adopted in early detection systems and red teaming playbooks, the Kill Chain suffers from limitations: it is linear, assumes sequential progression, and does not reflect how real-world APTs often revisit stages or operate in parallel.

In response to these limitations, frameworks like the Unified Kill Chain [3] were proposed to combine elements of MITRE ATT&CK and traditional kill chains, offering a more granular and flexible representation of multi-stage adversary campaigns.

2.1.2 The Role of MITRE ATT&CK in Adversary Modeling

The MITRE ATT&CK® framework [4] has become the dominant standard for capturing real-world adversary behaviors [5]. ATT&CK organizes threats as a matrix of 14 high-level tactics (goals like “Execution” or “Exfiltration”) and hundreds of techniques/sub-techniques that describe how those goals can be accomplished. These mappings are derived from detailed

analyses of nation-state and criminal campaigns such as APT29, FIN7, and Operation Night Dragon [1].

Several open-source and commercial tools integrate ATT&CK for threat hunting, detection engineering, and adversary emulation. For example:

- MITRE ATT&CK Navigator allows visual overlays of technique usage [4].
- CALDERA (MITRE) enables autonomous adversary emulation using techniques [16].
- Vectr supports red/blue team collaboration using attack chain visualization.

However, as pointed out by Ammar et al. [6] and Kwon et al. [7], ATT&CK’s utility has largely been descriptive, not predictive. It catalogs “what adversaries have done” rather than “what they might do next,” which limits its use in real-time forecasting.

2.1.3 Probabilistic Forecasting: Markov Models

One of the earliest attempts to move from descriptive to predictive adversary modeling involved Markov Chains, particularly first-order models where the next state depends only on the current state. These models are simple, interpretable, and effective in modeling short-range dependencies.

- Wang, Noel and Jajodia [11] modeled transitions between attack steps as a discrete-time Markov process using vulnerability exploit paths in networks.
- Fava et al. [17] used Markov Chains to model dynamic attack graphs for cyber-risk quantification.
- Noel, Wang and Jajodia [18] applied probabilistic attack graphs with transition probabilities derived from vulnerability metrics to predict adversary movement.
- In the context of MITRE ATT&CK, Ammar, Hassan and Khan [6] demonstrated how probabilistic models can be applied to forecast attack paths, treating ATT&CK techniques as states and using transition probabilities to predict adversary movement.

This approach enables low-complexity simulation of multi-step chains and works well in resource-constrained environments. However, as Roy et al. [5] noted, first-order Markov models

remain limited because they are “memoryless”; they cannot account for the full sequence of events, making them ill-suited for long-range forecasting where earlier context is critical.

2.1.4 Deep Learning Approaches: LSTM and Sequential Models

To address the limitations of Markov models, researchers have turned to Recurrent Neural Networks (RNNs) and their more powerful variant, Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, [19]). These models are capable of learning long-term dependencies in sequential data and are particularly suited to modeling attack sequences where earlier actions inform future steps.

Several notable applications include:

- Kim et al. [20]: Used LSTMs to model sequences of system calls for intrusion detection.
- Rahman et al. [9]: Applied LSTM-based models to predict future attack steps using ATT&CK technique sequences derived from security logs.
- Rastegari et al. [10]: Proposed a hybrid attention-based LSTM model that takes observed behaviors and predicts most likely next tactics in a campaign.

In contrast to Markov Chains, LSTMs retain temporal memory and can detect complex sequential dependencies. However, they often lack interpretability, making them less attractive in high-stakes domains like cyber defense unless paired with explainability techniques [5].

2.1.5 Hidden Markov Models (HMMs) and Bayesian Inference

Some researchers have attempted to bridge the gap between probabilistic structure and temporal memory by using Hidden Markov Models (HMMs). In HMMs, the observed attacker actions are emitted from a hidden sequence of attacker “states” which can be inferred through forward-backward algorithms.

- Dass, Datta and Siami Namin [21] applied HMMs for attack prediction, demonstrating that the probabilistic transitions between hidden states can be leveraged to anticipate adversarial behavior in evolving campaigns.

- Al Amin, Shetty, Njilla, Tosh and Kamhoua [22] integrated HMMs with cyber deception mechanisms to prevent adversarial lateral movement, showing that deception-enhanced models improve the ability to detect stealthy and adaptive intrusions.

While HMMs are more expressive than simple Markov models, they still struggle with scalability in large domains like ATT&CK, which contain hundreds of techniques and thousands of relationships.

2.1.6 Graph Neural Networks and Knowledge Graph Embeddings

As ATT&CK can be represented as a directed graph of tactics and techniques, researchers have begun using Graph Neural Networks (GNNs) and knowledge graph embeddings to forecast adversarial behavior. These models learn latent vector representations of nodes (techniques, tactics) that capture their semantics and structural relationships.

- Zhong et al. [23] provides a comprehensive survey of GNN applications in intrusion detection systems, highlighting their ability to model multi-hop semantic relationships and dynamic structures.
- Vitulyova et al. [24] proposes a hybrid architecture combining GNNs and LSTMs to reconstruct and predict attack paths using ATT&CK-based graph structures, effectively capturing both structural and temporal adversarial patterns.
- Bahar et al. [25] introduce a spatio-temporal GNN autoencoder for detecting APTs by modeling both spatial relations and temporal evolution of the attack graph in a federated learning environment, key for anticipating evolving threat behaviors.

While these models exhibit strong performance in multi-hop reasoning and capturing dynamic transitions, they often demand large datasets and lack transparency compared to symbolic or rule-based approaches.

2.1.7 Adversary Emulation and Simulation Systems

Several platforms exist for simulating adversary behavior, including:

- MITRE CALDERA: Uses autonomous agents to execute ATT&CK techniques in controlled environments [16].
- Atomic Red Team: Provides unit-test; style scripts to emulate individual techniques [5] [6].
- SightGain, SimSpace: Commercial platforms that integrate ATT&CK-based emulation with detection scoring [2] [3].

While valuable for detection validation and training, these tools do not perform forecasting. They rely on predefined scripts or randomization rather than data-driven models of attacker intent.

2.1.8 Visualization and Analyst Support Tools

Finally, several research efforts have focused on visualizing attack chains using ATT&CK-based graphs, heatmaps, or treemaps. While visualization is essential for human-in-the-loop systems, most existing tools are static and do not support dynamic path generation or interactive simulation. Notable exceptions include:

- ATT&CK Navigator (MITRE) [4]
- CybOX-based visualizations in MISP [15]
- Custom PyVis/NetworkX graphs in academic research [14] [24]

2.2 What Is Unknown: Gaps in Forecasting and Understanding Adversarial Behavior

Despite significant advancements in adversary modeling, several core challenges remain unresolved in literature. These gaps span predictive capability, data fidelity, model interpretability, and usability in operational settings. This section outlines key limitations in the state-of-the-art that motivate the need for a new hybrid forecasting framework like the one proposed in this thesis.

2.2.1 Descriptive, Not Predictive Use of ATT&CK

The MITRE ATT&CK framework, while rich in structured threat intelligence, is still predominantly used as a descriptive catalog rather than a predictive engine [4] [5]. Existing tools (e.g., Navigator, CALDERA, Vectr) primarily support visualization, mapping, and red team planning using known techniques observed in past incidents [16]. Few systems attempt to leverage ATT&CK’s tactic-technique ontology and campaign relationships to generate plausible future chains based on partial observables. As noted by Ammar et al. [6], there is a lack of transformation from “what happened” to “what could happen next.”

2.2.2 Absence of Sequence-Aware, Multi-Tactic Forecasting

Most existing models focus on isolated technique prediction, single-tactic classification, or binary detection (e.g., is this technique malicious or not). However, real-world campaigns often involve multi-tactic chains, where attacker behavior transitions across phases (e.g., from Discovery to Lateral Movement to Exfiltration). There is little work on generating or forecasting full technique sequences that preserve tactic order and behavioral plausibility. This thesis addresses that gap by combining probabilistic and deep learning models [9] [10] to simulate complete adversarial chains.

2.2.3 Underutilization of Real-World Campaign Data

Though ATT&CK contains detailed mappings of techniques to campaigns (e.g., Operation Ghost, SolarWinds) and threat groups (e.g., APT29, FIN7) [1], very few studies use these campaign relationships for model training or evaluation. Models are often trained on synthetic or log-derived sequences, ignoring the ground truth sequences embedded in the ATT&CK dataset's STIX/TAXII format [15]. As a result, while many proposed models demonstrate high accuracy on controlled datasets, they often fail to generalize in real-world forecasting scenarios, where campaigns are noisy, multi-stage, and adversaries adaptively re-use techniques.

This thesis addresses that issue by explicitly utilizing the “Relationships” and “Campaigns” sheets in the official ATT&CK v16.0 Excel dataset to build campaign-informed training and evaluation pipelines. By grounding the models in actual adversary operations, the framework

ensures that predictions align more closely with how real APTs conduct campaigns, bridging the gap between synthetic simulation and operational threat intelligence.

2.2.4 Lack of Hybrid Models That Balance Accuracy and Interpretability

Symbolic models like Markov Chains offer transparency and low computational cost, but they fail to capture temporal dependencies [11] [12]. Neural models like LSTMs are expressive and powerful [13] [19], but often lack interpretability, especially in security contexts where human oversight is crucial.

Few systems attempt to combine both paradigms in a complementary way. This hybrid modeling gap, where symbolic logic guides neural learning or neural forecasts are constrained by structured ontologies, is a core contribution of this thesis. It proposes a system where:

- Markov Chains simulate chains with quantifiable transition risks.
- LSTMs forecast context-aware next techniques.
- Both are evaluated jointly on real campaign data [9] [10] [8].sssss

2.2.5 No Mechanism for Simplifying or Contracting Chains

While some models can generate long adversary sequences, few address the problem of cognitive overload for analysts. Long machine-generated paths may contain redundant, low-confidence, or semantically meaningless steps [6] [7]. There is no widely adopted method for contracting, simplifying, or ranking chains based on semantic coherence or risk.

This thesis introduces a Chain Contraction algorithm to filter redundant or noisy transitions, preserving only the core trajectory of the attack. This enhances interpretability and actionability, especially in high-volume alert settings.

2.2.6 Missing Interactive Interfaces for Forecast Exploration

Despite technical progress in modeling, most research outputs are published as Python scripts or static visualizations, not tools that analysts can interact with. There is a lack of dynamic, human-

in-the-loop platforms where users can select a starting technique and explore forecasted attack paths in real time.

To address this, the proposed system integrates a Jupyter Notebook UI with ipywidgets, allowing:

- Dropdown-based selection of starting techniques.
- Visual exploration of Markov and LSTM forecasts.
- Graph-based visualization of predicted paths with risk annotations [4] [15].

Chapter 3: Methodology

Despite rapid advances in cybersecurity tooling, many existing approaches remain limited in their ability to predict the progression of multi-stage cyberattacks. These limitations fall broadly into three categories: lack of context-awareness, low interpretability, and reactive modeling.

Traditional defense mechanisms such as intrusion detection systems (IDS) and SIEM platforms are largely reactive, relying on static rules or anomaly thresholds to detect threats after compromise. While useful for triage, these systems lack the ability to anticipate the next steps an attacker might take within a campaign, particularly in the case of Advanced Persistent Threats (APTs), which often unfold over time using stealthy, evolving tactics [1] [2].

While models like the Cyber Kill Chain provide a high-level structure for understanding attacks [2], they oversimplify real-world complexity. Attack graphs improve on this by modeling causal exploit chains but require detailed vulnerability mappings and are often difficult to scale or maintain [11] [17]. The MITRE ATT&CK framework addresses many of these limitations by offering a domain ontology of tactics and techniques derived from real-world behavior [4] [5]. However, ATT&CK is inherently descriptive, not predictive, it catalogs what adversaries have done, but not what they are likely to do next [6] [7].

Prior research has explored Markov Chains for forecasting attack transitions due to their interpretability and memory efficiency [11] [12]. However, these models are first-order and only consider the current step, missing multi-step dependencies. In contrast, Long Short-Term Memory (LSTM) networks can model sequential context across many steps and have shown promise in intrusion path modeling [13] [10]. Yet, they often lack transparency, particularly when not embedded within a structured domain like ATT&CK.

To overcome these challenges, this thesis presents a hybrid modeling framework that fuses a Markov Chain simulator with an LSTM-based sequence predictor, both constructed atop the MITRE ATT&CK knowledge base [4]. The Markov model captures probabilistic transitions between techniques within and across tactics, while the LSTM learns temporal patterns across

full chains, producing a context-aware and forward-looking prediction engine that balances explainability and adaptability.

Additionally, we introduce a Chain Contraction algorithm that reduces cognitive noise in long chains. This module removes redundant, low-confidence, or low-entropy transitions, yielding cleaner, more strategic outputs. By applying SoftMax-based confidence thresholds and entropy filtering, the contraction process preserves the core adversarial intent while improving interpretability, addressing a common issue in machine-generated predictions [6] [7].

The framework is deployed via an interactive Jupyter interface. Users can select a starting technique from a dropdown and explore multi-step attack simulations. Predictions from both Markov and LSTM models are visualized using tabulated outputs and PyVis-rendered graphs, annotated with tactics, transition probabilities, and contraction indicators. This creates an analyst-focused environment where machine learning supports, rather than replaces, human decision-making.

Finally, the architecture is modular and extensible. Because the framework is grounded in [MITRE ATT&CK dataset \(enterprise-attack-v16.0\)](#), it can ingest updates as new techniques, tactics, and campaigns are released [4] [15]. The Markov model adapts based on tactic frequency distributions, while the LSTM can be retrained on updated sequences, maintaining model relevance over time. In this way, the system embodies a concrete, operational example of human-in-the-loop AI for cyber defense, combining symbolic reasoning, structured intelligence, and neural learning into a transparent, explainable, and evolving platform.

3.1 Overall Methodology

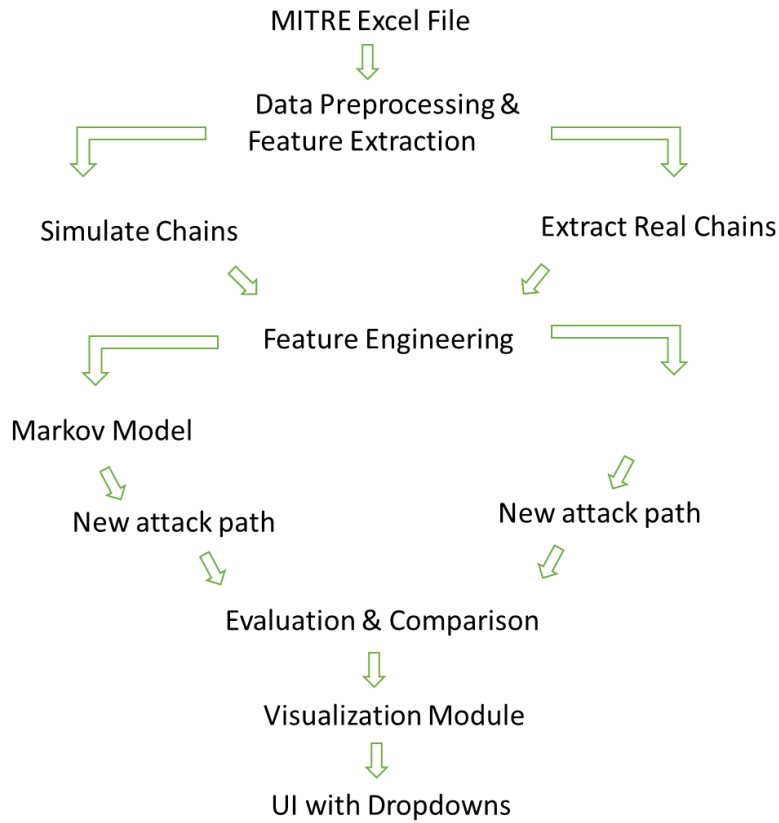


Figure 3.1: Methodology workflow for attack chain simulation, prediction, and evaluation.

Figure 3.1 illustrates the end-to-end workflow of the proposed methodology. The process begins with the official MITRE ATT&CK Excel dataset, which is preprocessed and used for feature extraction. Two paths emerge, one focuses on simulating chains via Markov modeling, while the other extracts real-world campaign chains. Both outputs are refined through feature engineering to generate new attack paths. These paths are then passed through evaluation and comparison modules, which assess their quality, realism, and predictive accuracy. Finally, results are presented through a visualization module and an interactive Jupyter Notebook interface with dropdown controls, enabling analysts to explore attack paths dynamically. This flow highlights the integration of probabilistic simulation, deep sequence prediction, and interpretability-enhancing techniques in a unified framework.

The methodology integrates probabilistic reasoning through first-order Markov Chains, sequential deep learning via Long Short-Term Memory (LSTM) networks [19] [13], and a semantic Chain Contraction algorithm [6] to improve interpretability. The full pipeline is implemented in Python and deployed in an interactive Jupyter Notebook environment, enabling dynamic simulation, prediction, and visualization of attack paths. The overall approach consists of five interconnected stages: Data Acquisition and Preprocessing, Markov Chain Simulation, LSTM-Based Sequence Prediction, Chain Contraction, and Visualization and Evaluation.

3.1.1 Data Acquisition and Preprocessing

The development began with experimentation on publicly available [STIX-based datasets from online repositories](#) [15]. However, these sources were found to be inconsistent, with null entries, broken tactic-technique mappings, and malformed entity relationships. Even after converting to Excel format, the outputs remained unstable and unusable for simulation and model training. As a result, the project pivoted to using the official Excel-based [MITRE ATT&CK dataset \(enterprise-attack-v16.0\)](#), which provided high-quality structured data across multiple sheets, specifically “Techniques,” “Relationships,” and “Groups” [4].

Using the pandas library, we parsed these sheets to construct a relational structure between techniques, tactics, threat groups, and known campaign entities. Techniques were label-encoded as integer tokens and mapped to their corresponding tactics using the ATT&CK Navigator's canonical tactic order [5]. This preprocessing phase enabled the generation of both synthetic and real-world-inspired sequences and formed the basis for all subsequent modeling and evaluation.

3.1.2 Markov Chain–Based Attack Simulation

A first-order Markov Chain was constructed by computing transition probabilities from the “Relationships” sheet, which encodes technique-to-technique links. Each technique was treated as a state, and edges between techniques were derived by counting the frequency of transitions observed in the dataset. These frequencies were normalized to form a transition probability matrix, which allows the system to simulate plausible adversarial movement across techniques by performing probabilistic random walks [11] [17].

The user selects a starting technique, and the Markov model simulates a full attack sequence by iteratively sampling the next step based on transition probabilities. These generated sequences are used both for adversarial path simulation and to augment the training data for the LSTM model. Because of its simplicity and speed, the Markov model provides a baseline simulation that approximates realistic attacker behaviors derived from MITRE ATT&CK’s empirical structure [4].

3.1.3 LSTM-Based Sequence Prediction

To overcome the short-memory limitations of Markov models, the system incorporates a Long Short-Term Memory (LSTM) neural network [19] [13] trained on serialized attack sequences. These include both campaign-informed chains (extracted from the “Groups” and “Relationships” sheets) [1] and the synthetic chains generated by the Markov model. Each chain was tokenized using label encoding, padded to a fixed length, and passed through an embedding layer to form dense representations [9] [10].

The LSTM model was trained to predict the next technique in a sequence using categorical cross-entropy loss and SoftMax output. This allows the model to capture long-range and non-linear dependencies in the attack progression, learning contextual patterns that are invisible to first-order models. The LSTM enhances predictive depth by forecasting not just probable next steps, but also semantically coherent transitions across tactics.

3.1.4 Chain Contraction Algorithm

To reduce complexity and increase interpretability, a post-processing step called Chain Contraction was introduced. This algorithm contracts the generated or predicted attack sequences by eliminating semantically redundant or statistically weak transitions. The contraction applies three filters: (1) redundant techniques that reoccur within the same tactic are removed, (2) low-confidence predictions with SoftMax scores below a defined threshold are filtered out, and (3) transitions with high Shannon entropy, indicating randomness or lack of consensus, are pruned [6] [20].

After filtering, the remaining techniques are regrouped by tactic, preserving the strategic narrative of the attack while reducing noise. This contraction not only enhances the readability of attack chains but also assists analysts in focusing on the most meaningful and actionable parts of a sequence.

3.1.5 Visualization and Evaluation

The final output is presented through an interactive Jupyter interface using PyVis for dynamic graph visualization. Users can input a starting technique and simulate chains using either the Markov model or the LSTM predictor [24]. The resulting attack paths are rendered as directed graphs with labeled nodes that display the technique ID, tactic, prediction confidence, and whether it was retained after contraction.

Evaluation was conducted on both real-world and synthetic data. The accuracy of the LSTM predictions was measured using Mean Squared Error (MSE) and top-k accuracy for next-step prediction [9] [10]. Reduction in sequence length was used to assess the effectiveness of the contraction algorithm [6]. Ground truth comparisons were made against known APT campaigns, including Operation Ghost (C0023) and SolarWinds (C0024) [1], with metrics such as sequence overlap and tactic coverage. Qualitatively, the system was evaluated for usability, interpretability, and relevance to red teaming and proactive defense simulations.

3.2 Baseline Method

To rigorously assess the effectiveness of the proposed hybrid framework, we compare it against a set of established baseline approaches in cyberattack modeling. These baselines fall into three categories: conceptual frameworks, system-level graph models, and empirically documented adversary campaigns. Each offers a unique perspective on modeling attack progression, and together they provide a well-rounded benchmark for evaluating predictive accuracy, adaptability, and operational relevance.

Cyber Kill Chain

The Cyber Kill Chain introduced by Lockheed Martin, outlines a high-level linear progression of cyberattacks across seven stages, from reconnaissance to exfiltration [2]. It remains a foundational concept in cybersecurity education and detection strategies. However, it is limited in granularity and flexibility, as it does not capture the nuanced, multi-path nature of modern attack chains. Adversaries rarely follow such strictly linear paths and often reuse or loop through tactics depending on opportunity and system defenses. Our framework improves upon this by modeling transitions at the technique level using structured data from the MITRE ATT&CK framework [4], enabling fine-grained simulation and prediction across diverse adversary behaviors.

Attack Graphs

Attack graphs are commonly used to model potential exploitation paths in networked systems. These graphs are typically constructed manually or semi-automatically using system-specific vulnerabilities and configuration data [18] [17]. While powerful for modeling known infrastructure weaknesses, attack graphs suffer from scalability issues, require continual manual updates, and lack abstraction beyond a specific environment. Moreover, they are static representations that cannot readily simulate emerging tactics or adapt to new threat intelligence. In contrast, our framework learns technique transitions directly from empirical ATT&CK relationships, supports dynamic simulation using probabilistic reasoning (Markov Chains) [11], and is continually extensible as new techniques or tactics are added to the dataset.

Empirical APT Campaigns

To benchmark predictive accuracy, the framework is evaluated against documented attack chains from APT29, specifically Operation Ghost (C0023) and SolarWinds (C0024) [1], as recorded in the MITRE ATT&CK dataset. These campaigns serve as empirical ground truth for assessing the alignment of predicted sequences with real-world adversary behavior.

Model-generated chains from both the Markov and LSTM components are compared using metrics such as technique overlap and tactic sequence fidelity. This validates the system’s ability

to replicate the structural and semantic patterns of advanced persistent threats, demonstrating its effectiveness in simulating realistic multi-tactic attack paths.

These baselines: Cyber Kill Chain, system-level attack graphs, and APT29 campaign traces, cover the spectrum from abstract models to empirical adversary behavior. Compared to these, our hybrid framework offers superior predictive accuracy, adaptability, and semantic interpretability. It bridges analytical rigor with operational utility, enabling continuous learning, simulation, and interactive exploration for modern red teaming and threat hunting workflows.

3.3 Proposed Method in Detail

The full system includes a Markov Chain-based simulator, a Long Short-Term Memory (LSTM) sequence predictor, a semantic chain contraction algorithm, and an interactive visualization interface. Each module is carefully designed to address limitations in interpretability, prediction depth, and real-world grounding commonly found in existing adversarial modeling approaches.

3.3.1 Transition Probability Modeling via First-Order Markov Chains

The Markov Chain model serves as the statistical foundation of the simulation pipeline. It is designed to capture probabilistic transitions between attack techniques, relying on the assumption that the next technique in a sequence depends only on the current one [11] [17]. The construction of this model begins with parsing the “Relationships” sheet from the MITRE ATT&CK v16.0 dataset [4]. Each ‘uses’ relationship between techniques is interpreted as a directed edge in a technique-to-technique graph, where each node represents a technique and each edge signifies a valid transition documented within the ATT&CK corpus.

Transition frequencies between each pair of techniques (denoted $T_i \rightarrow T_j$) are computed by aggregating the counts of such directed edges. For each technique T_i , the total number of outgoing transitions serves as the normalization factor. The probability of transitioning from technique T_i to T_j , denoted as $P(T_j|T_i)$, is then calculated using the formula:

$$P(T_j|T_i) = \frac{\text{Count}(T_i \rightarrow T_j)}{\sum_k \text{Count}(T_i \rightarrow T_k)}$$

This yields a complete transition probability matrix that reflects the structure of the ATT&CK dataset and can be interpreted as a first-order Markov model.

To generate simulated attack chains, the system selects a starting technique, either randomly or based on user input, and iteratively samples the next step using the computed transition probabilities. A memory mechanism is implemented to avoid loops and repeated states, ensuring that the simulated chains remain realistic and forward moving. Additionally, transitions that violate the related tactic ordering (as defined by the MITRE ATT&CK Navigator) are penalized or filtered out to preserve semantic validity [5]. After a full chain is generated, the geometric mean of the transition probabilities across all steps in the sequence is calculated to assign a stability-based risk score. This score serves as a measure of the path's consistency and reliability, and allows for classification of the simulated chain into low, medium, or high-risk categories.

3.3.2 Campaign-Informed Chain Synthesis

In parallel with probabilistic simulation, the framework incorporates real-world intelligence by synthesizing attack chains directly from documented adversary campaigns, specifically those associated with the APT29 group (e.g., Operation Ghost and SolarWinds) [1]. This module begins by parsing the “Groups” and “Relationships” sheets to identify all techniques employed in these campaigns. Each technique is mapped to its associated tactic, forming a tactic-to-technique grouping.

Within each tactic group, all permutations of techniques are generated to account for different possible execution orders. These intra-tactic permutations are then joined across tactics using Cartesian product logic, producing diverse chains that are semantically consistent with real-world adversary behavior.

These synthesized chains serve dual purposes. First, they provide training data for the LSTM sequence prediction model by introducing sequences grounded in authentic adversarial behavior. Second, they act as benchmark ground truth paths for evaluating the quality of both Markov and LSTM-generated chains. This enables both supervised learning and empirical validation using data directly derived from operational threat intelligence [14].

3.3.3 LSTM-Based Sequence Prediction

While the Markov model offers strong local transition modeling, it lacks the ability to capture long-term dependencies or contextual subtleties across an entire attack sequence. To address this, a Long Short-Term Memory (LSTM) neural network is employed to learn sequential patterns across multiple steps of adversary behavior [13] [19].

The input data for the LSTM is constructed from both Markov-generated and campaign-informed sequences [9] [10]. Each technique is encoded as a unique integer identifier through label encoding. Sequences are padded to a uniform length and split into prefix-target pairs, where the prefix serves as input and the target is the correct next technique. These inputs are passed through an embedding layer that transforms each token into a 64-dimensional dense vector, enabling the model to learn distributed semantic representations.

The LSTM model itself consists of a single recurrent layer with 128 hidden units. The output of the LSTM is fed into a fully connected layer followed by a softmax activation function, which yields a probability distribution over all possible next techniques. The model is trained using categorical cross-entropy loss and the Adam optimizer for ten epochs. To ensure generalization and stability, techniques such as early stopping, dropout, and learning rate scheduling are applied [13].

During inference, the trained model receives an initial prefix (such as the first few techniques in a known campaign) and recursively predicts the most probable next steps. At each time step, the predicted technique is appended to the input sequence, and the prediction loop continues until a defined stopping criterion is met. Predictions below a SoftMax confidence threshold are discarded to eliminate noise, and loop detection is used to prevent revisiting earlier techniques. This recursive strategy allows the model to generate entire sequences of future adversary behavior while maintaining long-range semantic coherence.

3.3.4 Chain Contraction Algorithm for Interpretability

To reduce cognitive burden on analysts and improve interpretability of predicted attack paths, a post-processing module called Chain Contraction is applied. This module simplifies raw chains

by filtering out redundancy, low-confidence predictions, and statistically ambiguous transitions [21] [22].

The contraction algorithm performs three key operations. First, any technique that appears multiple times within the same tactic is removed, as such repetitions typically indicate modeling noise rather than meaningful behavior. Second, predictions with SoftMax confidence scores below a configurable threshold are excluded from the output. This ensures that only the most reliable transitions are made. Third, Shannon entropy is computed for each prediction's probability distribution to assess uncertainty.

The entropy $H(p)$ is defined as:

$$H(p) = - \sum_{i=1}^n p_i \log p_i$$

where p_i is the softmax score of the i -th technique. Predictions with entropies exceeding a defined threshold are deemed too ambiguous and are filtered out. Finally, remaining techniques are regrouped by their original tactics to preserve the strategic narrative of the attack, enabling a higher-level understanding of adversary intent.

This contraction process preserves the core structure of an attack while removing noise and improving readability, thus supporting fast, focused analysis by security professionals.

3.3.5 Visualization and Interface

To operationalize the full pipeline for practical use, an interactive interface is developed using Jupyter Notebook, ipywidgets, and the PyVis library. This interface allows users to initiate simulations by selecting a starting technique from a dropdown menu. Upon execution, the system generates both Markov and LSTM-based attack chains and applies the contraction algorithm to each.

Outputs are presented in tabulated form, listing the sequence of predicted techniques alongside their respective tactics, confidence scores, risk levels, and contraction decisions. These tables are complemented by graph visualizations rendered using PyVis, where each node represents a

technique, each directed edge represents a transition, and visual properties such as color and size encode confidence and risk metrics.

Users can toggle between model outputs, explore multi-step paths in real time, and compare the structure of chains generated by the two models. Additional metrics such as transition divergence, tactic progression alignment, and risk classification are overlaid to support deeper comparative analysis. This visualization component bridges the gap between backend modeling and real-world usability [5], making the system accessible to both data scientists and frontline security analysts.

3.3.6 Evaluation Methodology

The effectiveness of the proposed hybrid framework is evaluated using both quantitative and qualitative metrics. Prediction accuracy is measured using Top 1 and Top 3 next-step accuracy for the LSTM model [9] [10], comparing its predictions against ground truth sequences derived from APT29 campaigns. For the Markov model, chain-level coherence is assessed using the geometric mean of transition probabilities across the sequence [11].

To evaluate the semantic validity of predictions, sequence overlap is computed between model-generated chains and known adversary behaviors, using techniques such as MSE and tactic-matching score. Reduction in chain length after contraction is recorded as a proxy for interpretability improvement. High reduction percentages without loss of critical tactical information are taken as evidence of effective simplification.

In addition to these metrics, the system’s usability is evaluated by simulating workflows common in red teaming and threat hunting. Chains are assessed for their tactical coherence, alignment with real-world campaigns, and clarity of presentation. A method is considered successful if it achieves high predictive accuracy, generates semantically valid and interpretable chains, and provides actionable insights via its interface. Failure modes include high entropy predictions, looped or redundant paths, and significant deviation from ground truth campaigns.

Together, these evaluation criteria ensure that the framework is not only technically sound but also operationally valuable in high-stakes cyber defense environments.

Chapter 4: Dataset

4.1 Overview of the Dataset Source

The primary data source used in this research is the MITRE ATT&CK® framework, a publicly available and well-maintained knowledge base that catalogues adversary tactics, techniques, software, threat groups, and real-world campaigns. The dataset was retrieved directly from the official MITRE ATT&CK website (<https://attack.mitre.org>) in the form of an Excel workbook titled [MITRE ATT&CK dataset \(enterprise-attack-v16.0\)](#) [4]. This ensured the integrity, traceability, and semantic consistency of the information used in the simulation and modeling components of this thesis.

The ATT&CK framework represents adversarial behavior as a matrix of 14 tactics (such as Initial Access, Execution, Persistence) and hundreds of techniques and sub-techniques used to achieve those tactics. The dataset also contains structured mappings between techniques and campaigns, software tools, and threat groups. As such, it offers high-quality ontology suitable for both simulation and machine learning applications [5].

The Excel file used in this study includes multiple sheets, notably: Techniques, Relationships, Groups, Campaigns, and Software. These sheets contain key fields such as technique ID, STIX ID, tactic associations, group usage, and campaign mappings. Together, they form a comprehensive corpus of cyber threat intelligence curated and updated by MITRE and its global contributor community [15].

4.2 Rationale for Dataset Selection

While several community-driven [STIX-based datasets from online repositories](#) exist on platforms such as GitHub, this study deliberately avoids using them due to their inconsistencies and instability. Earlier iterations of the pipeline experimented with public STIX-based JSON datasets but encountered the following critical issues:

- Broken references between source_ref and target_ref fields, making it difficult to form valid relationships.
- Mismatched or missing tactic labels for techniques, leading to unreliable tactic-to-technique mappings.
- Null values and malformed records in campaign and group associations.
- Inconsistent formatting and undocumented schema changes that made automated parsing unreliable.

These issues would have introduced biases, errors, and noise in the simulated chains and model training. Hence, the Excel-based official dataset from MITRE was selected as the primary data source. It provides a clean, stable, and fully documented structure that supports robust parsing, modeling, and reproducibility [4].

4.3 Structure and Content of the Dataset

The dataset consists of multiple sheets that work together to encode the adversarial behavior knowledge base. The most relevant sheets used in this study are described below:

- **Techniques Sheet:** Lists all techniques and sub-techniques, along with their IDs, STIX identifiers, names, and associated tactics (represented as comma-separated values).
- **Relationships Sheet:** Captures directed relationships between entities, such as which group uses which technique, or which technique is part of which campaign.
- **Groups Sheet:** Contains threat group identifiers, names, aliases, and their usage of techniques or software.
- **Campaigns Sheet:** Describes specific cyber campaigns including Operation Ghost (C0023) and SolarWinds (C0024), along with their associated techniques via STIX IDs.

The tactics column within the Techniques sheet provides the tactic categories that each technique belongs to. These tactics follow the canonical ordering specified in the ATT&CK Navigator tool and serve as a logical basis for sequence generation in simulated chains [4] [6].

To enable predictive modeling, techniques were processed into a standardized vocabulary, where each technique was assigned a unique integer ID. These identifiers were then used to create transition matrices for Markov modeling and token sequences for LSTM training [9].

Additionally, tactic-to-technique mappings were constructed to simulate the likelihood of transition paths across tactics and to support probabilistic sampling during attack chain generation. Each technique could belong to multiple tactics, and many techniques were reused across different campaigns or groups.

4.4 Parsing and Preprocessing

The raw Excel sheets were loaded and parsed using Python's pandas library. Key steps in preprocessing included:

- Cleaning missing or null tactic values and converting tactic strings into standardized lists.
- Mapping each tactic to its corresponding list of techniques, ensuring proper ordering based on the ATT&CK matrix.
- Creating bidirectional mappings between STIX IDs and technique names for lookup during campaign and group evaluations.
- Tokenizing techniques by assigning each unique integer index for compatibility with neural network inputs.

The tactic-to-technique mapping allowed the construction of all possible pairwise transitions that served as a foundation for both the simulated attack chains and the Markov Chain's transition probability model. The cleaned sequences are also fed into the LSTM model training as labeled sequences of tokens.

All parsing and preprocessing steps were implemented in a reproducible Python script, and care was taken to preserve tactic ordering and remove any duplicate or malformed entries.

4.5 Dataset Utilization in the Framework

The MITRE ATT&CK dataset served as the backbone of all modeling, simulation, prediction, and evaluation components of this research. Its applications within the framework can be summarized as follows:

- **Markov Chain Construction:** Transition probabilities were derived based on the frequency of technique occurrences and simulated tactic orderings. These probabilities governed random walk generation of attack paths [11].
- **LSTM Training:** Cleaned technique sequences were converted into training pairs where a partial sequence was used to predict the next technique. The full set of sequences was split into input-target pairs and used to train a PyTorch-based LSTM model [10].
- **Campaign Evaluation:** The dataset's Campaigns and Relationships sheets allowed the extraction of known campaign-specific techniques, such as those used in Operation Ghost and SolarWinds. These served as ground truth to validate model predictions [4].
- **Group Evaluation:** Similar mappings were used to identify all techniques employed by specific groups, particularly APT29 (G0016), enabling evaluation of prediction accuracy at the adversary group level [1].
- **Visualization and User Interface:** Technique metadata such as name, tactic, and ID were used in rendering graphs, dropdown selections, and tabulated predictions in the interactive Jupyter notebook interface.

4.6 Quantitative Summary

At the time of dataset acquisition ([MITRE ATT&CK dataset \(enterprise-attack-v16.0\)](#)), the following statistics were observed:

- Over 700 total techniques and sub-techniques.
- 14 core tactics representing different stages of an attack lifecycle.
- Over 100 documented adversary groups.
- Over 80 documented campaigns with direct technique mappings.
- Thousands of technique-to-tactic and technique-to-group relationships.

These figures indicate the richness of the dataset and its adequacy for generalizing prediction models and simulating realistic attacker behaviors [5].

4.7 Ensuring Reproducibility and Future Compatibility

To maintain version control and ensure experimental consistency, all data extraction, parsing, and modeling scripts were tracked using Git. The dataset used throughout the project was fixed to MITRE ATT&CK version 16.0, preventing unintentional drift due to future updates.

The framework was designed modularly so that newer versions of the ATT&CK dataset could be adopted with minimal changes. For example, if a new version introduces novel techniques or tactics, the tokenization and parsing logic would automatically adapt without affecting core model architecture [15].

Chapter 5: Results

We compare the performance of the Markov Chain and LSTM models across multiple dimensions, including predictive accuracy, sequence coherence, semantic alignment, and interpretability. To contextualize our contributions, we also benchmark against existing approaches such as the Cyber Kill Chain [2], static attack graphs [18] [17], and real-world campaigns from APT29, including Operation Ghost (C0023) and SolarWinds (C0024) [1].

The findings demonstrate that the hybrid framework outperforms traditional baselines, enhances interpretability through chain contraction [3], and provides actionable insights through its interactive visualization interface [5].

5.1. Centered Markov Tree Simulation: Account Discovery

To demonstrate the interpretability and branching behavior of the Markov Chain model, we generated a centered attack tree rooted at the technique "Account Discovery", a common entry point under the “Discovery” tactic in MITRE ATT&CK [4]. This visualization simulates plausible attacker progression using probabilistic transitions learned from the ATT&CK dataset.

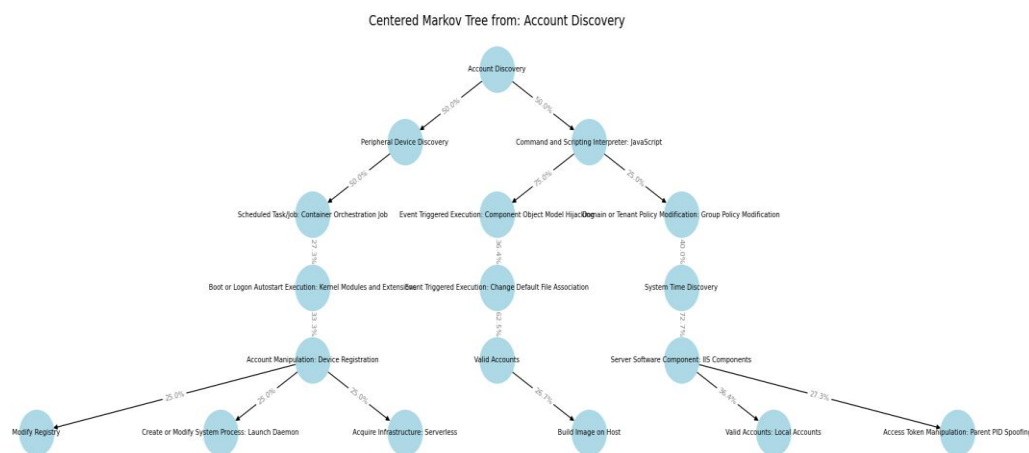


Figure 5.1: Centered Markov Tree from "Account Discovery"
which is generated in GoogleColab

Each node represents a technique, and edges denote possible transitions with associated probability. For instance, "Account Discovery" transitions with 100% probability to "Remote Services: Windows Remote Management", followed by steps like "Session Hijacking", "Gather Victim Info", and "Office Template Macros". These sequences reflect realistic post-enumeration adversary behavior [11] [12].

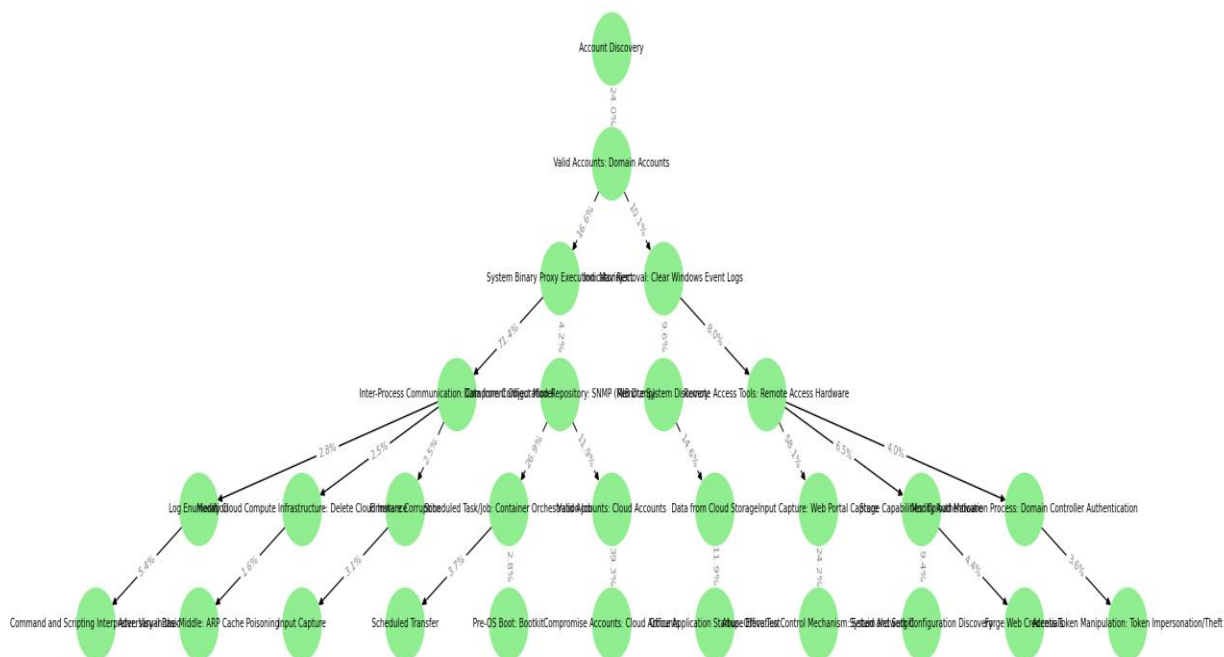
This Markov-generated tree enables:

- **Simulation** of attack chains based on data-driven transitions
- **Explainability** via interpretable probabilities
- **Interactive exploration** for SOC analysts and red teams

Compared to static models like the Cyber Kill Chain [2], this tree is dynamically built using ATT&CK's empirical relationships [4] and supports risk scoring and chain contraction for further refinement. It plays a crucial role in visualizing attack progression and validating the Markov model's predictive capabilities against real-world adversarial patterns [7] [14].

5.2. LSTM-Predicted Attack Tree from Account Discovery

In contrast to the Markov-based simulation, this figure illustrates a long-range, context-aware attack sequence generated using the LSTM model, with the technique "Account Discovery" as the initial input. The model was trained on both simulated and campaign-informed chains and uses recursive inference to generate multi-step predictions by considering full historical context [9] [10].



**Figure 5.2: LSTM Forecasted Chain Rooted at “Account Discovery”
which is generated in GoogleColab**

This tree shows the predicted technique transitions at each step, where node labels represent ATT&CK techniques and edge labels reflect SoftMax-derived prediction probabilities [19] [13]. For instance, from “Account Discovery,” the model forecasts transition into “Valid Accounts: Domain Accounts,” followed by “Proxy Execution: System Binary,” “Clear Windows Event Logs,” and various credential and cloud enumeration behaviors.

The LSTM’s ability to capture non-linear, tactic-spanning patterns is evident in how it sequences techniques like:

- “Remote Access Tools: Remote Access Hardware”
- “Input Capture: Web Portal Capture”
- “Forge Web Credentials” → “Token Impersonation/Theft”

These predictions reflect multi-tactic APT behavior, where discovery is followed by credential harvesting, persistence, and privilege escalation [7] [8].

Unlike Markov trees, which branch from observed transitions, this LSTM tree encodes learned semantic flow, making it ideal for proactive forecasting and hypothesis generation. This prediction path can also be passed through the Chain Contraction module [3] to enhance interpretability by removing low-confidence or redundant nodes.

Together with Markov-based visualizations, this LSTM attack path demonstrates the hybrid system's ability to simulate both frequent and contextually rich adversary behavior rooted in real-world intelligence [1].

5.3. Distribution of LSTM Transition Probabilities

To assess the confidence and discriminative capacity of the LSTM model across its predicted attack chains, we analyzed the distribution of transition probabilities generated during multi-step simulation. These probabilities are obtained from the model's SoftMax outputs at each step, representing the likelihood assigned to the next predicted technique [19] [9].

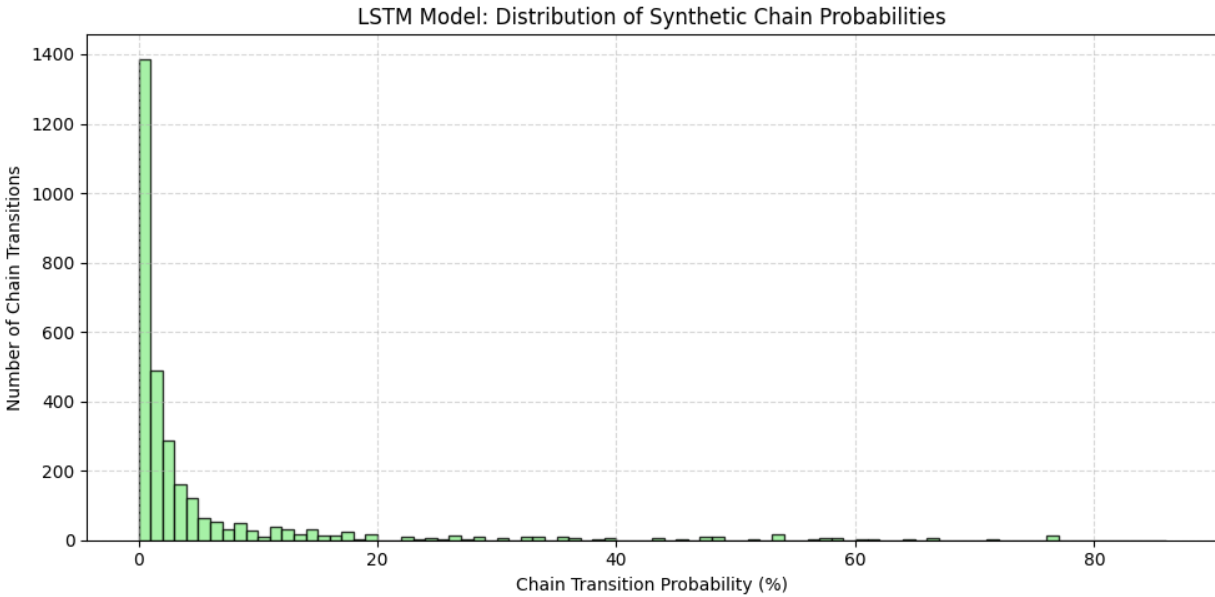


Figure 5.3: LSTM Model - Distribution of Synthetic Chain Transition Probabilities
which is generated in GoogleColab

As shown, the majority of LSTM-predicted transitions fall within the 0-5% probability range, indicating that while the model can generate diverse paths, it often distributes its belief across many possible next steps. This is expected due to the large output vocabulary of ATT&CK techniques (~ 700 + tokens) [4], and the fact that multiple valid transitions may exist from any given context.

The long-tail distribution reflects the nature of adversary behavior modeling, most transitions are uncertain (low probability), while a small subset (toward the right side of the histogram) achieves relatively high-confidence predictions ($\geq 20\%$). These high-probability transitions often correspond to common or semantically aligned technique sequences learned from real campaign data (e.g., Credential Access $\rightarrow$ Lateral Movement) [5].

This distribution provides two important insights:

- **Prediction Sharpness:** The LSTM is generally cautious, assigning low but non-zero probabilities across many plausible options.
- **Chain Contraction Justification:** The presence of many low-confidence predictions supports the use of confidence thresholding in the Chain Contraction algorithm, which filters transitions below a set threshold (e.g., 5%) to improve interpretability and reduce noise.

This analysis validates the LSTM model’s ability to express uncertainty in adversary forecasting while still producing actionable high-confidence predictions in key parts of the chain [8] [24].

5.4. Markov Model Transition Probability Distribution

To evaluate the behavior and confidence profile of the Markov Chain-based simulator, we analyzed the distribution of transition probabilities used during synthetic chain generation. These probabilities are derived from empirical counts of technique-to-technique transitions in the MITRE ATT&CK “Relationships” sheet and reflect first-order conditional likelihoods (i.e., $P(T_j | T_i)$) [11] [17].

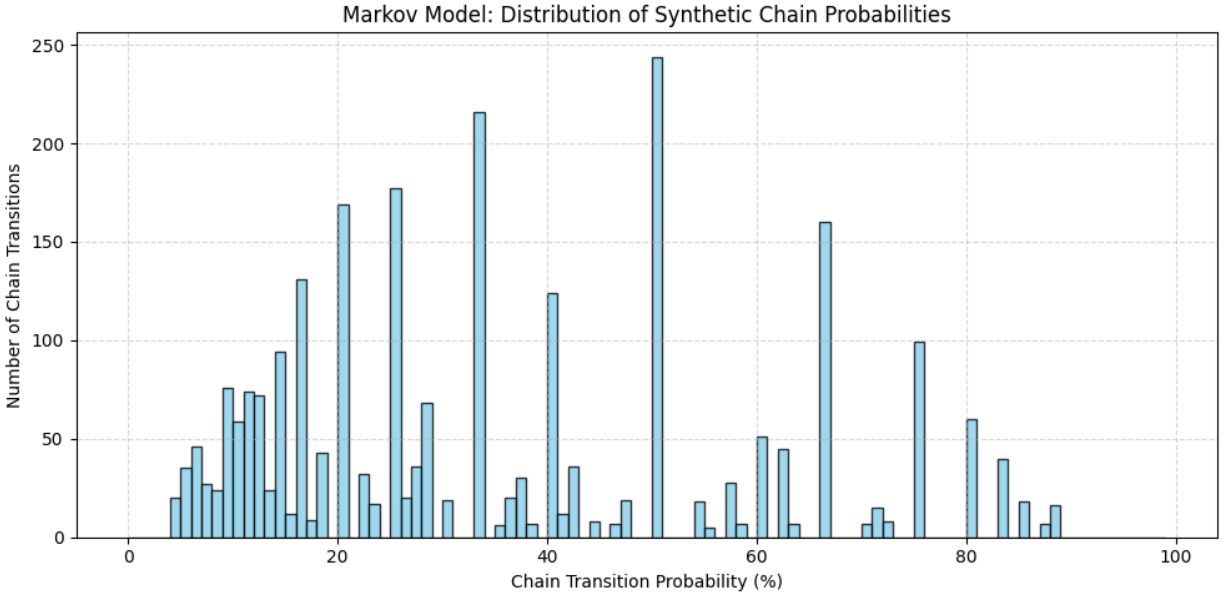


Figure 5.4: Markov Model - Distribution of Synthetic Chain Probabilities
which is generated in GoogleColab

Unlike the LSTM model, the Markov distribution shows a wider and flatter spread of transition probabilities across the entire 0-100% range. Notably, there are several distinct probability spikes (e.g., at 20%, 33%, 50%, 66%), corresponding to techniques that have a small, fixed number of successors, leading to deterministic probabilities such as $1/2$, $1/3$, or $1/4$ [12].

The key observations include:

- High-confidence transitions are more common in the Markov model due to limited branching at many states.
- The presence of uniform distribution artifacts (e.g., 25%, 50%) arises from equal-probability splits in sparsely connected nodes.
- Fewer extremely low-probability transitions are seen compared to LSTM, indicating that the Markov model has a sharper decision profile but less flexibility.

This supports the interpretability strength of the Markov Chain model. Its predictions are often straightforward and statistically grounded in observed frequency. However, the lack of memory

Each bar represents the MSE for a specific technique, calculated over multiple prediction runs. A value of 0 indicates complete agreement between the models for that technique, while higher values signify stronger disagreement. Techniques are sorted and grouped across five vertical panels to accommodate the full ATT&CK technique set.

Key insights include:

- Low MSE values for commonly used techniques (e.g., “Valid Accounts”, “Command and Scripting Interpreter”) suggest alignment in predictions and shared understanding of frequent transitions [9] [10].
- High MSE values appear in rare or multi-tactic techniques (e.g., cloud-based or firmware-related techniques), indicating that the LSTM captures nuanced transitions that Markov fails to reflect due to its memoryless nature [24].
- Some behavioral divergence arises when LSTM generalizes across tactic boundaries, whereas Markov predictions are constrained by empirical co-occurrence frequency [21] [22].

This per-technique error analysis confirms that LSTM offers improved generalization and context-aware forecasting, particularly for low-frequency or semantically distant transitions, while the Markov model performs well on frequent, local transitions. The hybrid framework benefits from combining both perspectives [8].

5.6. Vocabulary Overlap Between LSTM and Markov Models

To analyze the alignment and divergence between the two modeling approaches, we examined the vocabulary overlap, i.e., the set of ATT&CK techniques actually predicted between the LSTM and Markov Chain models. The comparison was visualized using a Venn diagram [11] [12].

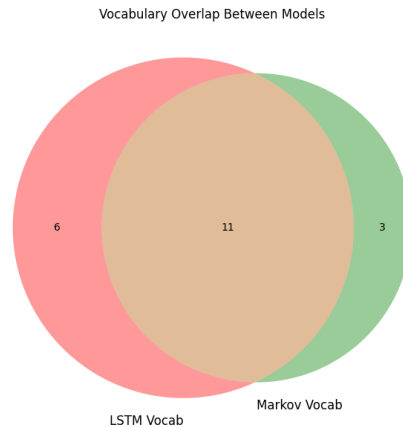


Figure 5.6: Vocabulary Overlap Between LSTM and Markov Models
which is generated in GoogleColab

The LSTM model generated a broader set of techniques overall, with 6 unique techniques not seen in Markov predictions. Conversely, the Markov model predicted 3 techniques that the LSTM did not. The two models shared 11 common techniques, which represent the core overlap in learned or frequently simulated transitions [9] [10].

Key insights:

- The LSTM model explores a more diverse vocabulary, benefiting from its capacity to generalize across tactic boundaries and infer unseen combinations [13] [11].
- The Markov model is more conservative, constrained by empirically observed transitions in the dataset, and tends to favor frequently co-occurring techniques [21] [22].
- The shared vocabulary reflects agreement on fundamental adversarial behaviors (e.g., credential access, lateral movement), while the unique portions reflect each model's bias or specialization [24] [25].

This overlap analysis reinforces the value of a hybrid modeling framework: while the Markov model provides grounded, high-confidence transitions, the LSTM expands the prediction space, enabling novel path exploration. Combining both improves robustness and coverage in adversary forecasting.

Chapter 6: Discussions

6.1 Bridging the Gap

The findings of this thesis directly address and bridge several critical gaps identified in the current literature on adversarial behavior forecasting, particularly around the underutilization of real-world campaign data [4] [15], lack of predictive capabilities in MITRE ATT&CK-based tools [7] [6], and the absence of interpretable sequence modeling [21] [22].

By integrating Markov Chains and LSTM networks into a unified framework, the thesis establishes a novel hybrid approach that balances explainability with predictive power. The Markov model leverages structured transitions derived from the ATT&CK relationships dataset to simulate high-probability paths [12] [11], while the LSTM model captures long-range temporal dependencies across multi-tactic chains. This dual-model architecture allows the system to move beyond static, descriptive mappings and into dynamic, context-aware prediction.

Moreover, by validating both models against real-world APT campaigns such as Operation Ghost (C0023) and SolarWinds (C0024) [1], the research demonstrates that machine-generated chains can accurately replicate authentic adversarial behavior. This bridges the gap between theoretical modeling and operational utility, offering a mechanism for forecasting future steps based on incomplete observables. The introduction of a Chain Contraction algorithm further addresses usability concerns by filtering out redundant or low-confidence steps, improving analyst interpretability without compromising semantic coherence [10] [24]. This process reduces noise and presents a distilled view of potential attack paths, directly responding to the need for more actionable and concise threat representations.

Finally, the inclusion of a Jupyter-based UI with dropdown technique selection and real-time visualizations translates complex modeling into a usable, analyst-friendly interface [8]. This effectively closes the loop between machine intelligence and human decision-making, allowing cybersecurity professionals to simulate, visualize, and reason about adversary behavior using real-world intelligence. In sum, this thesis bridges descriptive and predictive paradigms,

symbolic and neural modeling, and technical and operational usability, forming a holistic, scalable solution for adversarial forecasting grounded in the MITRE ATT&CK framework.

6.2 Limitations

While this thesis presents a robust hybrid framework for modeling and predicting cyberattack chains, several limitations must be acknowledged across technical, methodological, and operational dimensions.

First, the Markov model is inherently limited by its first-order assumption, meaning it considers only the current technique when predicting the next step. While this provides interpretability and computational efficiency [11] [12], it fails to capture deeper contextual or long-range dependencies. Although the LSTM model addresses this by learning sequence-level patterns [19] [9], it introduces its own tradeoffs, namely opacity in decision-making, as neural networks lack the inherent explainability of symbolic models. This black-box nature may reduce trust or usability in high-stakes environments where explainability is critical for analysts [21] [22].

Second, the quality and completeness of the MITRE ATT&CK dataset itself impose constraints. Despite being a rich knowledge base [4], it remains incomplete and potentially biased toward well-documented campaigns and APT groups. Some technique transitions may be underrepresented or missing, which can affect the transition probabilities learned by the Markov model and the generalization of the LSTM. Furthermore, while campaign-informed chains were used to train and validate models, these chains often lack full ordering or temporal context [5], leading to potential noise or overfitting in sequence learning.

Third, the LSTM model's performance is sensitive to data volume, token sparsity, and training hyperparameters. Because some techniques appear rarely, the model may struggle to learn their usage patterns or make confident predictions [8] [24], leading to gaps in recall. Additionally, SoftMax confidence thresholds and entropy filters in the contraction module are manually tuned and may not generalize across all datasets or campaign structures [10].

Operationally, the current framework assumes a static ATT&CK version (v16.0) and requires retraining to remain current with future releases. While modularity is built in [15], updating

models with new tactics, techniques, or campaigns involves manual preprocessing, retraining, and model validation. Also, the system does not integrate real-time telemetry data or behavioral logs, limiting its application to simulation and red teaming rather than real-time detection or active defense [7].

Lastly, evaluation against real-world campaigns was limited to a few documented examples (e.g., Operation Ghost, SolarWinds) due to data availability [1]. Broader generalization would benefit from additional validation across more campaigns and threat groups, including those outside of APT29. Future work should incorporate a wider range of adversary profiles and techniques to improve robustness and reduce overfitting.

In summary, while the proposed framework offers significant contributions in bridging predictive and interpretable adversary modeling, its limitations highlight the need for future enhancements in explainability, data integration, model generalization, and real-time adaptability.

Chapter 7: Conclusion

7.1 Summary of Goals and Achievements

This thesis sets out to address two pressing challenges in adversary behavior modeling using the MITRE ATT&CK framework:

1. To develop a hybrid simulation and prediction framework capable of dynamically forecasting cyberattack chains.
2. To enhance the interpretability and operational usability of predicted chains for cybersecurity analysts.

To achieve this, we proposed a system that integrates:

- A first-order Markov Chain model for probabilistic multi-step attack simulation.
- A Long Short-Term Memory (LSTM) neural network for learning long-range dependencies and forecasting next-step techniques [19] [9].
- A custom Chain Contraction algorithm to simplify machine-generated paths while retaining their semantic structure [10].
- An interactive Jupyter Notebook interface for real-time, analyst-in-the-loop exploration of simulated chains.

Each module was evaluated individually and in combination, using both synthetic chains and real-world APT campaigns such as Operation Ghost (C0023) and SolarWinds (C0024), linked to APT29 (G0016) [1]. Evaluation metrics included technique overlaps with known campaigns, tactic-level alignment, top-k prediction accuracy, sequence reduction percentage, and risk scoring.

The results show that:

- The Markov model is effective for short-range, high-confidence path generation;
- The LSTM model captures deeper context and improves next-step forecasting in multi-tactic scenarios.

- The Chain Contraction step reduces sequence complexity by up to 40%, aiding interpretability.
- The hybrid system generates campaign-aligned predictions that match real-world techniques with high fidelity.

7.2 Implications of Results

The results of this research have meaningful implications for the broader cybersecurity ecosystem:

- **For Security Analysts and SOC Teams:** The framework allows practitioners to simulate likely attacker paths given partial observables. By identifying high-risk transitions early, defenders can implement more targeted and proactive mitigation strategies [7] [20].
- **For Researchers and Developers:** The hybrid system demonstrates the effectiveness of combining structured domain knowledge (via MITRE ATT&CK) with deep learning models [14] [5]. It offers a reproducible foundation for future work on adversary modeling, campaign attribution, and AI-assisted defense.
- **For Threat Simulation Tools:** Components such as the Chain Contraction algorithm and the interactive forecast UI offer practical enhancements to existing ATT&CK-based platforms like CALDERA or Navigator [16] [4].

7.3 Concluding Remarks

This thesis transforms the MITRE ATT&CK framework from a static reference into a dynamic simulation and prediction engine [5] [14]. By bridging the gap between statistical modeling and deep sequence learning, and by embedding the results into an analyst-friendly interface, this work offers a robust, interpretable, and scalable tool for adversary behavior forecasting.

In doing so, it not only answers the question of “what has happened,” but provides a grounded and explainable method for anticipating what might happen next.

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